

Khush Shah

Carnegie Mellon University | B.S. in Computer Science & Mathematics | Expected May 2029
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Fully work authorized | U.S. Citizen

EDUCATION

Carnegie Mellon University

B.S. in Computer Science & Mathematics

Pittsburgh, PA

Expected May 2029

- **Relevant Classes:** Deep Learning (Graduate Level), Machine Learning, Data Structures & Algorithms, Competition Programming, Computer Systems, Linear Algebra, Multivariable Calculus, Differential Equations

EXPERIENCE

Maximor | *Software Engineer (Part-Time)*

August 2026 – Present

Software Engineer Intern

June 2026 – August 2026

- Designed and deployed the company's first multi-agent system on Claude Managed Agents: a coordinator agent delegating to subagents via cross-thread tool-call routing, replacing separate Temporal workflows
- Revised the production fleet of Claude Managed Agents, rewriting system prompts, tightening tool schemas, reworking input framing, and debugging live agent runs to improve reliability and output quality
- Designed the human-in-the-loop clarification system used across all agents: an answer/skip/rewind question lifecycle, durable persistence of agent runs and sessions, and a reviewer-facing UI
- Wrote the generic managed-agent runtime powering it: pausing live agent sessions at clarification tool calls, resuming them via Temporal, and staging session files to balance context delivery with performance
- Currently designing the company's agent evaluation harness: regression evals gating the migration of live customer workloads to newer models, plus orchestration tooling to run agents reproducibly
- Served as a forward-deployed engineer, shipping customer-requested features and bug fixes across the product

Carnegie AI Safety Initiative (CASI)

September 2025 – Present

- Studied transformers in depth through a technical AI safety reading group, covering mechanistic interpretability, RLHF, alignment techniques, and the fundamentals of modern ML research on LLM internals
- Co-leading a 3-person research project on AI safety alignment robustness, producing empirical research

ScottyLabs Tech Team | *CMU Courses Backend*

September 2025 – June 2026

- Built automated Rust system to query CMU APIs and recursively discover related course resources
- Wrote Python ETL to normalize course data into Postgres and expose it via JSON and CSV endpoints
- Rebuilt the web app in TypeScript + React using TanStack Router; platform used by 1000s of CMU students

RESEARCH

Geometric Framework for Predicting Fragility of LLM Safety Training Methods

April 2026 – Present

- Using RepE-extracted concept vectors and Hessian eigenvector analysis to show safety concepts occupy lower-rank, higher-curvature subspaces than capability directions
- Showing that benign fine-tuning updates preferentially project onto fragile safety directions, with representational rank and layer depth as independent fragility predictors
- Building a framework that predicts DPO- vs PPO-trained models' alignment robustness from geometric properties, validating on HarmBench across Llama, Mistral, and Qwen; targeting ICLR 2027 workshops

LEADERSHIP EXPERIENCE

Los Altos Hacks | *Organizer & Sponsorship Lead*

2021 – 2025

- Organized the world's largest high-school hackathon; personally raised \$20,000+ in sponsorships; gave speeches to 1000+ students and mentored many new members

Computer Engineers of the Next Generation (CENG) | *Senior VP*

2022 – 2025

- Managed 100+ volunteers teaching 24+ coding classes (Python, Java, JavaScript) to underserved schools; created over \$1M of impact; earned the *Presidential Volunteer Service Award* for 200+ service hours in a single year

TECHNICAL SKILLS

ML: Claude Managed Agents, LLM APIs, MCP; implemented transformers from scratch; completed ARENA AI safety curriculum; working knowledge of LLM internals, attention, RLHF, and alignment techniques

Languages: Python, Java, C++, C, C#, JavaScript, TypeScript, Rust, SQL, CSS, Standard ML, $\text{\LaTeX}$

Tools & Platforms: Git, Linux, Docker, PostgreSQL, Snowflake, Temporal, Firebase, MongoDB, GCP, AWS

Libraries & Frameworks: React, Node.js, FastAPI, PyTorch, TensorFlow, NumPy, pandas, scikit-learn